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FS2 AERO SAFETY OF NAVIGATION AERONAUTICAL PRODUCTION SYSTEM (APS) ADDENDUM

NATIONAL GEOSPATIAL-INTELLIGENCE AGENCY
AERONAUTICAL NAVIGATION OFFICE

VERSION 1.0

28 APR 21

1 (U) Background

(U) This Addendum provides system specific information and constraints related to support of the Aeronautical Production System (APS) under the Foundation Sustainment Services (FS2) Aeronautical Safety of Navigation (SoN) Task Order.

(U) The APS environment (SoN, VO and MIDB) utilizes and includes COTS, GOTS, FOSS, Custom Code software/applications, numerous tools, output/ingest applications/tools and multiple iterations of the databases (Aeronautical, Vertical Obstructions, Production, AFD, JPIP, MIDB) both spatial and relational to perform the missions.

(U) APS's databases currently operate on the NGA high side environment from traditional virtual servers and within the Amazon C2S environment (RDS/E2). Future plans, captured in the Solution Epic Source Content Management Intelligence Capability Baseline Description (ICBD) include an APS environment in the UC2S. APS's Acquisition and Dissemination activities include all three NGA Networks, the WWW, GHUB, FVEY, COI's and Contractor sites.

2 (U) Work Requirements

(U) The overall scope includes (but is not limited to) Program Management, (WBS 1.0) Operations Support, (WBS 5.0) and Software Maintenance (WBS 4.0) of the APS capabilities.

(U) The APS production system is currently undergoing a major technology overhaul from outdated/unsupported software and infrastructure to modern technologies and processes. Despite this upgrade, the current system provides little additional capability beyond basic performance increases. In order to take advantage of the technology and capabilities available, there are required changes to established processes and the corresponding components of APS. This step is a necessary to enable future capabilities and meet upcoming requirements (i.e., DAFIF 9, AIXM and PBN). All APS processes shall comply and be in conformance with:

- ISO 9001:2015 certification
- Communication Navigation and Surveillance/Air Traffic Management (CNS/ATM) DO-200B/FAA AC-153B
- Standards to maintain the SFA FAA/CNS/ATM TYPE 1 Data Provider LOA.
- Extensible to support future needs.

3 (U) Personnel and Skills Requirements for APS

(U) The contractor shall provide personnel for Places of Performance located at the NGA West Campus and at the Contractor Facility.

(U) The contractor shall provide the appropriately cleared personnel onsite at NCW four (4) FTE: two (2) Database Administrators, one (1) DevOps Engineer and one (1) Software Engineer to perform the contracted functions in order to access Aero SoN databases that contain Classified and Unclassified/LIMDIS data.

(U) The contractor shall have expertise skill sets critical to restoring NGA Mission Critical Aero SoN APS capabilities with the documented NGA Continuity Planning Tool (CPT) restoration timelines.

(U) The Contractor shall have experience with APS, without it can dramatically impact operations and production processes of NGA's Aeronautical Safety of Navigation Office (SFA).

(U) The Contractor must possess experience with and knowledge of AIRAC production cycles and NGA Aeronautical production processes.

(U) The Contractor shall identify and provide one primary and one alternate person to perform the set-up and execution of all software integrations and deployments in-line with the NGA prescribed dev/ops pipeline. This is inclusive of all locations (domain, network, etc.) that the software is deployed to for development, user testing, and operations. These individuals shall be appropriately cleared to perform the required tasks on all domains that they are required.

(U) The Contractor shall have working knowledge of the current NGA SFA Aeronautical processes and procedures in order to provide for an uninterrupted transition into future capabilities and requirements.

(U) The Contractor shall be knowledgeable with the following APS Data Base(s), GUI's, Applications (AAS and AOE), Ingest/Generation Software, Data Extraction/Export Capabilities, and Software/Applications Quality Routines and Tools Utilized in Operations

- Aeronautical (Postgres)*, Vertical Obstructions (Postgres), Production (SyBase), AFD, Tasking, JPIP, MIDB* and maintenance of all the relational and spatial capabilities of the Databases, BCP requirements, Quality checks, Quality scripts, DB replication, data maintenance, associated data integrity/quality checks, associated documentation (CONOPS, SRD, DB Schemas, Security accreditations, etc.), database modeling and management.
- Aeronautical (AAS), BEAR, NOTAM Query, JPIP, AIMS, AFD and Vertical Obstructions and software baselines (Aeronautical, JPIP, AIMS, AFD and Vertical Obstructions).
- AFD, VO (Pre-Compare, Target Data ingest, ISQ NOTAMS, etc.) NOTAM, BOOKSHELF (eNFDD), NACO, Terminals, Co-Production, JPIP (AFLD), JPIP (VO), FAA AIXM, AIXM, and XML.
- DAFIF, AAFIF, DCDT, AACDTD, DVOF/WebDVOF (.shp, VPF, VVOD, GML, KML, KMZ, CHUM/OCF, Powerline extraction) , AFD, TFADS (VO), TFADS (Aeronautical), EAREP, FIF, Co-DAFIF, FLIP DVD, Terminals, Co-Pro, AOE outputs, AQP.
- AFD, VO, NOTAM, BOOKSHELF (NFDD), JPIP (AFLD), JPIP (VO), MIDB, .ul files, DB files, FLIP, TAPS, E-PIL, GTO, BCP requirements, Text files, FAA files.
- MAGVAR check/update, DVC, BRC, FIF, RAV, TAPS, VO Elevation service, VO Pre-Compare, DB De-Classify, CRC, Tasking, TAPS, Integrity/quality and Athena.

(U) The contractor shall provide technical expertise and on-demand backup capabilities to accomplish work in the following task areas listed below:

Information Technology Service Management
Technology Assessment & Evaluation
Systems Engineering
Testing and Verification
Project Management and Planning
Configuration Management
Tech Writing and Documentation Support
Customer Engagement and Operations Planning
Software Engineering, Development and Integration

Mission and Business Application/Tool Development, Integration and Maintenance
Web and Portal Systems Development, Integration, Maintenance and Management
Knowledge and Content Management
Maintenance & Test Environment
Data Services, Data Administration, and Database Management
Enterprise Data Backup, Disaster Recovery (DR), and Continuity of Operations (COOP) Operations and Support
Cybersecurity and Information Assurance Services
Cloud (AWS/CS2) Architecture, Management, Security and function

4 (U) APS Specific Constraints / System Information

(U)The Contractor shall provide capacity to provision and maintain the development, integration, testing, test environments, operation environments, tools, applications, security requirements, and software baselines for APS on the network(s) where it resides. (WBS 1.0, 3.31, and 5.31)

4.1 (U) Software Support Services Specific Constraints (WBS 4.0)

(U) The Contractor shall adhere to the Adaptive Maintenance broad range of activities listed below. (WBS 4.17)

- Small and simple tasks (e.g. processing a new or modified data type from existing source)
- Moderately complex tasks (e.g., incorporating a new collector type) to
- Extensive and highly complex changes requiring modification of the APS system, components or application design, source code and/or COTS modification, and testing (regression testing on all deliverables, verification, validation, and security). The extensive/complex changes require close coordination with users, other contractors, COTS vendors, and external customers. Examples, of the larger and highly complex tasking include redesign of critical interfaces (e.g. MIDB/MCAT), Database major version upgrades, significant code re-writes due to security-driven code base updates, etc.), adapting a system or application to a new platform (hardware, and/or operating systems), transitioning to/from cloud environments or to different cloud providers), and incorporating a new phenomenology.

4.2 (U) Operations Support Services Specific Constraints (WBS 5.0)

(U) The Contractor shall provide Agile operations with a Release/Sprint schedule of <u>2-week sprints, 4 week releases</u> - coordinated with production cycles. APS Output Production schedules (includes, but is not limited to):

- DAFIF-Every 28 days,
- AAFIF-Weekly,
- DCDT-Weekly,
- MIDB-Twice Daily,
- AFD-Daily,
- Aeronautical and Vertical Obstruction Database/Digital File Outputs-Weekly,
- Integrity/Quality Checks- Twice Daily,
- Vertical Obstructions: (DVOF/WebDVOF, VVOD, OCF, AACDTD, etc.) Every 28 days
- AQP: Every 28 days
- FLIP DVD: every 28 days.

(U) The contractor shall follow the Agile processes in regards to Sprint reviews, TEM's and other required meetings.

(U) The Contractor shall deliver and deploy for testing and operations, utilizing the "Jenkins pipeline" (current process) for deployment activities: software upgrades, patches, modified source code, and revised documentation.

(U) The Contractor shall include all components required for installation along with all documentation required to demonstrate compliance with NGA's security and accreditation processes as applied to the environment(s) where the software will be deployed.

(U) The Contractor shall ensure all software utilized within the APS environment(s) be maintained to within one major version of the most current version as prescribed by the NGA SWAP listings and cloud/domain environment(s) (Software Addendum) requirements unless security or operational requirements dictate otherwise. The final decision for deviations belong with the Aeronautical Safety of Navigation Office (SFA).

(U) The contractor shall ensure all software delivery packages contain updated software, source code, HP Fortify results (.fpr file and .pdf file), appropriate documentation (i.e., release notes, change logs, test results, installation instructions, build sheets, etc.) and any additional components (software, scripts, documentation, etc.) required for deployment.

(U) The contractor shall ensure the scans for HP Fortify must be run against the most current rule set, contain no hits from the OWSAP top ten, and have all false-positive hits addressed.

(U) The contractor shall ensure that no less than one year would be required for an ASP transition to gain the requisite expertise with the system without significant oversite from SFA. (WBS 1.3)

(U) The contractor shall provide a transition plan that assures no service interruption and no interruptions to the SFA's production schedules. (WBS 5.41)

(U) The contractor shall provide Operations Support services for the APS environment in a cloud-based Production/Integration/Test/Operations environments, utilizing COTS, GOTS, FOSS, Custom Code software/applications, numerous tools, output/ingest applications/tools, and multiple iterations of the databases (Aeronautical, Vertical Obstructions, Production, AFD, JPIP, MIDB) both relational and spatial to perform the mission.

(U) The contractor shall have the capability to operate in the NGA Top Secret environment for the current APS Databases.

(U) The contractor shall be able to utilize all three NGA Networks, the WWW, GHUB, FVEY, COI's and Contractor sites for APS Acquisition and Dissemination activities.

(U) APPENDIX A: ACRONYMS

AACDTD	Area Arrival Chart Depicting Terrain Data
AAFIF	Automated Air Facilities Intelligence File
AAS	Aeronautical Application Suite
AFD	Airfield Foundation Data
AIMS	Aeronautical Information Management System
AIRAC	Aeronautical Information Regulation and Control
AIXM	Aeronautical Information Exchange Model
AOE	Aeronautical Obstruction Environment
APS	Aeronautical Production System
AQP	Airfield Qualification Program
ASP	Application Service Provider
AWS	Amazon Web Service
BCP	Bulk Copy Program
BEAR	Bearing and Range Tool
BRC	Bearing/Range Calculator
C2S	AWS Region (Classified)
CHUM	Chart Update Manual
CNS/ATM	Communication Navigation Surveillance/Air Traffic Management
COI	Community of Interest
CONOPS	Concept of Operations
COTS	Commercial Off The Shelf
CRC	Cyclic Redundancy Check
DAFIF	Digital Aeronautical Flight Information File
DB	DataBase
DCDT	Digital Chart Data Transfer
DVC	Data Validation Check
DVOF	Digital Vertical Obstruction File
E-IPL	Electronic Instrument Procedure Library
EAREP	Expanded Airport Report
EC2	Elastic Cloud Computing
FAA	Federal Aviation Agency
FIF	Flight Information File
FLIP	Flight Information Publication
FOSS	Free Open Source Software
FVEY	An International partnership consisting of five nations
GHUB	Geospatial Data Repository
GML	Geography Markup Language
GOTS	Government-Off-the-Shelf
GTO	Geospatial Terminal Operations
GUI	Graphic User Interface
IAVA	Information Assurance Vulnerability Assessment
ICBD	Intelligence Capability Baseline Description

INC	Incident
ISO	International Organization of Standards
JPIP	JPEG 2000 Interactive Protocol
KML	Keyhole Markup Language
KMZ	Keyhole Markup Zipfile
LIMDIS	Limited Distribution
LOA	Letter of Authorization
MAGVAR	Magnetic Variation
MIDB	Modernized Integrated Database
NACO	National Charting Organization
NFDD	National Flight Data Digest
NGA	National Geospatial-Intelligence Agency
NOTAM	Notice To Airmen
O&M	Operations Maintenance
O&S	Operations Support
OCF	Obstruction/Obstacle Change File
OWSAP	Open Web Application Security Project
PaaS	Platform as a Service
PBI	Problem Based Initiative
PBN	Performance Based Navigation
.pdf	File format
RAV	Range of Allowable Values
RDS	Relational Database Service
RFC	Request for Change
S&S	Sustainment Support
SFA	NGA's Aeronautical Safety of Navigation Office
.shp	Shapefile format
SOW	Statement of Work
SoN	Safety of Navigation
SRD	System Requirements Document
SWAP	Software Approval Process
TAPS	Terminal Approach Production System
TEM	Technical Exchange Meeting
TFADS	Table Formatted Aeronautical Data Set
UC2S	AWS Region (Unclassified)
ul	Unload files
VO	Vertical Obstruction
VPF	Vector Product Format
VVOD	Vector Vertical Obstruction Data
WWW	World Wide Web
XML	Extensible Markup Language